

Langlands for SL2

Intuition here

1 Build up

Let us start with some useful results from representation theory:

Theorem 1.1: Clifford's Theorem

G be a group, $N \trianglelefteq G$ a normal subgroup, K a field, $\pi : G \rightarrow \mathrm{GL}_n(K)$ be an irreducible representation. Then the restriction $\pi|_N$ breaks up into a direct sum of irreducible representations of N of equal dimensions (i.e. it is semisimple and each representation is of the same dimension). These irreducible representations of N lie in one orbit for the action of G by conjugation on the equivalence classes of irreducible representations of N (i.e. they are isotypic components with the same multiplicity).

Proof :

Let W be an irreducible subrepresentation of $V|_H$. Since H is normal in G , for $g \in G$, H acts irreducibly on gW by $(h, gw) \mapsto hgh^{-1}hw$. The subspace $\sum_{g \in G} gW$ is a nonzero subrepresentation of V . Since V is irreducible, it is equal to V . Since a representation which is a sum of irreducible sub-representations is semisimple, $V|_H$ is semisimple. A similar argument follows for the 2nd assertion, completing the proof.

Thus, assume $H \trianglelefteq G$ is a finite index normal subgroup, and let π be an irreducible K -representation of H . Let us show that it is a component of an K -representation Π of G whose restriction to H contains π . First, induct so that we get $\mathrm{Ind}_H^G \pi$. This is finite length still as $[G : H] < \infty$, and contains an irreducible representation, say Π . Then there is a non-trivial map $\Pi \rightarrow \mathrm{Ind}_H^G \pi$. By Frobenius:

$$\mathrm{Hom}_G(\Pi, \mathrm{Ind}_H^G \pi) \cong \mathrm{Hom}_H(\Pi|_H, \pi)$$

Hence, as π is irreducible and the restriction is semisimple, π must be in the direct summands of $\Pi|_H$. Next, if χ is an K -character of G that is trivial on H , the restriction of $\Pi \otimes \chi$ to H contains π , for

$$(\Pi \otimes \chi)|_H \cong \Pi|_H$$

If there is another irreducible K -representation Π' that contains π , it is a twist of Π :

Lemma 1.2: lemma 1

Let K algebraically closed and G/H abelian. Then Any irreducible K -representation Π' of G containing π is isomorphic to $\Pi \otimes \chi$ for some K -character χ of G trivial on H .

Proof :

Certainly $\text{Hom}_H(\Pi'|_H, \Pi|_H) \neq 0$. The right adjoint of the restriction from G to H is the induction Ind_H^G from H to G , therefore Π' is isomorphic to an irreducible subrepresentation of $\text{Ind}_H^G(\Pi|_H)$. We have $\text{Ind}_H^G(\Pi|_H) \simeq (\text{Ind}_H^G 1) \otimes \Pi$ because G/H is finite, and the irreducible subquotients of $\text{Ind}_H^G 1$ are the characters χ of G trivial on H because K is algebraically closed. Therefore, there exists χ such that $\Pi' \simeq \Pi \otimes \chi$, completing the proof.

Suppose H is a closed subgroup of a locally profinite group G and V a K -representation of G . If $[G : H] < \infty$, H is open, and conversely if H is open and cocompact the index is finite. If V is smooth, $V|_H$ is smooth and if H is open and $V|_H$ is smooth the V is smooth.

Let V be irreducible and $V|_H$ have finite length. Let W be an irreducible component of $V|_H$, π an isomorphism class, and G_π the G -stabilizer of π . Let V_π be the π -isotypic component of $V|_H$. Then the G -stabilizer of V_π is G_π . The G -stabilizer of W is open in G as it contains the nonzero G -stabilizer of $v \in W$ and V is smooth, and furthermore it is contained in G_π . Both of these groups have finite index in G and by Clifford's theory $V = \text{Ind}_{G_\pi}^G(V_\pi)$. The representation G_π on V_π is irreducible and the length of $V|_H$ is:

$$\text{len}(V|_H) = [G : G_\pi] \text{len}(V_\pi|_H)$$

Lemma 1.3: lemma 2

Assume that G/H is abelian. Then:

1. G_π is normal in G and does not depend on the choice of π in $V|_H$. The smooth K -characters of G trivial on G_π are in X_V .
2. Assume K algebraically closed.
 - (a) Any irreducible subquotient of the smooth induction $\text{Ind}_H^G 1$ is a smooth K -character χ of G trivial on H .
 - (b) Any irreducible K -representation of G containing π is a twist $V \otimes \chi$ of V by some smooth K -character χ of G trivial on H .
 - (c) When $V|_H$ has multiplicity 1, then $W = V_\pi$, for a smooth K -character χ of G trivial on H , $V \otimes \chi \simeq V$ if and only if χ is trivial on G_π , and G_π is the largest subgroup I of G containing H such that $\text{lg}(V|_I) = \text{lg}(V|_H)$.
 - (d) When K is algebraically closed and $V|_H$ has multiplicity 1, then

$$|X_V| = \begin{cases} [G : G_\pi] & \text{if } \text{char}_K = 0, \\ [G : G_{\pi, \ell}] & \text{if } \text{char}_K = \ell > 0, \end{cases}$$

where $G_{\pi, \ell}$ is the smallest subgroup of G containing G_π such that $[G : G_{\pi, \ell}]$ is relatively prime to ℓ .

Proof :

1. The isotypic components of $\Pi|_H$ are G -conjugate, their G -stabilizers are G -conjugate and contain H hence they are equal because G/H is abelian. Since $V \otimes \chi \simeq \text{Ind}_{G_\pi}^G(\chi|_{G_\pi} \otimes V_\pi)$ for any smooth K -character χ of G , the smooth K -characters of G trivial on $G(\pi)$ are in X_V .
2. (a) For any closed subgroup Q of G and a smooth K -representation X of Q , the represen-

tation $\text{Ind}_Q^G X$ is the space of functions $f : G \rightarrow X$ with the property $f(qgk) = qf(g)$ for $q \in Q$, $g \in G$, $k \in K_f$ for some open subgroup K_f of G , with the action of G by right translation, and where $\text{ind}_Q^G 1$ is the subrepresentation on the subspace of functions of compact support modulo Q . When G/Q is compact, $\text{Ind}_Q^G X = \text{ind}_Q^G X$. Let $V \supset U$ be G -stable subspaces with V/U irreducible. We can suppose V generated by an element f (indeed $V'/U' \simeq V/U$ for the G -stable space V' generated by $f \in V \setminus U$ and the kernel U' of the map $V' \rightarrow V/U$). There is an open subgroup K of G which fixes f . We have $U \subset V \subset \text{ind}_K^G 1$ and one is reduced to the case where G/H is finite.

- (b) The proof of Lemma ?? remains valid with the smooth induction Ind_H^G , which is the smooth compact induction $\text{ind}_H^G 1$, because G/H is compact, so that $\text{ind}_H^G(\Pi|_H) = \Pi \otimes \text{ind}_H^G 1$.

3. Any smooth character χ of G trivial on H with $\text{ind}_{G_\pi}^G(V_\pi) \simeq \text{ind}_{G_\pi}^G(V_\pi \otimes \chi|_{G_\pi})$ is trivial on G_π . Indeed, restricting to G_π we see that $V_\pi \otimes \chi|_{G_\pi}$ is conjugate to V_π by some $g \in G$. Restricting to H gives that $\pi \simeq \pi^g$, so $g \in G_\pi$, hence $V_\pi \otimes \chi|_{G_\pi} \simeq V_\pi$. As $\text{Ker}(\chi)$ is open in G and G/H is compact, $J = \text{Ker}(\chi) \cap G_\pi$ has finite index in G_π . If χ is not trivial on G_π then the action of J on V_π is reducible. Indeed, $\text{ind}_J^{G_\pi}(1)$ contains subrepresentations 1 and $\chi|_{G_\pi}$, and by Frobenius reciprocity $\text{End}_J(V_\pi|_J)$ is equal to $\text{Hom}_{G_\pi}(V_\pi, \text{ind}_J^{G_\pi}(V_\pi|_J)) = \text{Hom}_{G_\pi}(V_\pi, V_\pi \otimes \text{ind}_J^{G_\pi}(1))$.

Hence $\dim(\text{End}_J(V_\pi|_J)) \geq 2$ and $V_\pi|_J$ is reducible. By the hypothesis of multiplicity 1, $V_\pi|_H$ is irreducible, hence $V_\pi|_J$ is irreducible as $H \subset J$. So χ is trivial on G_π .

The group G_π is a subgroup I of G containing H with $\text{len}(V|_I) = \text{len}(V|_H)$. If I has this property, the restriction to H of any irreducible component on $V|_I$ is irreducible, hence I is contained in G_π .

4. follows from (3).

1.4 Restriction and Multiplicity Assume that $V|_H$ has multiplicity 1. The G -stabilizer of any irreducible component of V is G_π . Denote $G_\pi = G_V$. Let I be a subgroup of G containing H . The number of orbits of I in the irreducible components of $V|_{G_V}$ is $\text{len}(V|_I)$. This number is the same for I and IG_V , hence $\text{len}(V|_I) = \text{len}(V|_{IG_V})$. We deduce that $G_V \subset I$ if $V|_I$ is reducible and $|G/I|$ is a prime number.

Lemma 1.5: lemma 3

Let θ be a smooth K -representation of a closed subgroup $U \leq H$. Then if the functor $\text{Hom}_U(-, \theta)$ is exact on smooth K -representations of H and $\dim \text{Hom}_U(V, \theta) = 1$, then $V|_H$ has multiplicity 1.

Proof :

We denote by $m_V(\pi)$ the multiplicity of any irreducible smooth K -representation π of H in $V|_H$. By exactness,

$$\sum_{\pi} m_V(\pi) \dim \text{Hom}_U(\pi, \theta) = \dim \text{Hom}_U(V, \theta) = 1.$$

There is a single π with $m_V(\pi) = \dim \text{Hom}_U(V, \theta) = 1$.

2 p -adic Reductive Group

Need theorem 3.2 and corollary 3.5

Let G be a p -adic (algebraic) reductive group, that is the group of F -rational points of a reductive connected F -group $\underline{G}$, so $G = \underline{G}(F)$.

Insure to incorporate this: https://en.wikipedia.org/wiki/Reductive_group

Theorem 2.1

Let $f : \underline{H} \rightarrow \underline{G}$ be an F -morphism of reductive connected F -groups such that $f(H)$ is a normal subgroup of G of compact quotient $G/f(H)$. Then:

1. The restriction to $f(H)$ of any irreducible admissible K -representation of G is semisimple of finite length.
2. Any irreducible admissible K -representation of $f(H)$ is contained in some irreducible admissible K -representation of G restricted to $f(H)$, and extends to an irreducible admissible representation of some open subgroup of G of finite index.

Proof :

G satisfies (2-1) and $f(H)$ satisfies the property (2-2) because H does. Apply the results of Section 2.2.

2.2 Remark 3.4 The condition that f is central in [Proposition 3.3](#) is necessary. Indeed, assume $\text{char}_F = 2$ and $f : \underline{\text{GL}}_2 \rightarrow \underline{\text{SL}}_2$, $f(g) = \phi(g)/\det g$ where $\phi(x) = x^2$ for $x \in F$ is the Frobenius.⁴ The F -morphism f is surjective but not central. Let $G = \text{GL}_2(F)$, $G' = \text{SL}_2(F)$, T' the diagonal torus of G' and U the group of unipotent upper triangular matrices in G' . Then $f(G) = T'\phi(G')$ is closed but not normal and not cocompact in G' (since $\phi(U) = U \cap T'\phi(G')$ and $U/\phi(U)$ homeomorphic to F/F^2 is not compact).

Corollary 2.3

Assume K algebraically closed. Let $f : \underline{H} \rightarrow \underline{G}$ be an F -morphism of connected reductive F -groups which induces a central F -isogeny $\underline{H}^{\text{der}} \rightarrow \underline{G}^{\text{der}}$ between the derived groups. Then the conclusions of [Theorem 3.2](#) apply to $f(H)$.

Proof :

The F -isogeny $\underline{H}^{\text{der}} \rightarrow \underline{G}^{\text{der}}$ is surjective with finite kernel contained in the centre of $\underline{H}^{\text{der}}$ [[Springer 1998, §12.2.6](#)]. If $\underline{Z}$ is the connected centre of $\underline{G}$, the natural map $\underline{Z} \times \underline{G}^{\text{der}} \rightarrow \underline{G}$ is surjective [[Springer 1998, Corollary 8.1.6](#)]. Hence the obvious map $\underline{Z} \times \underline{H} \rightarrow \underline{G}$ is surjective and central. [Proposition 3.3](#) applies to $Zf(H)$. But K being algebraically closed, Z acts by a character in any irreducible smooth K -representations of G , and we get the corollary.

2.4 Remark 3.6 There is a more elementary proof that the restriction to $f(H)$ of any irreducible admissible K -representation of G is semisimple of finite length in [[Silberger 1979](#)].

3 Restriction of $GL_2(F)$ to $SL_2(F)$

Let F be a locally compact nonarchimedean field with residue characteristic (ex. $F/\mathbb{Q}_p$), and $L = \bar{L}$ an algebraically closed field of characteristic $\neq p$. This paper will be about K -representations of $SL_2(F)$. The reader can assume $K = \mathbb{C}$ to link this to the intuition complex representations like in [1]. The goal is to show how the Local Langlands correspondence goes from being a bijection to a surjection, with fibers of size 1, 2, or 4; these fibers are called *L-packets* or *L-indistinguishable's*.

Let $G = GL_2(F)$, $G' = SL_2(F)$, Z the center, G the torus, B the Borel subgroup, U the unipotent subgroup. If $X \subseteq G$ is a subset, $X' = X \cap G'$ and $x = (x_{ij})$ represents a matrix in G

Take $H = ZG'$. It is a closed normal compact group where G/H is isomorphic to $F^*/(F^*)^2$. It is normal as Z is obviously normal and $G' = \ker \det$ where $\text{im } \det = F^*$, and $gZG'g^{-1} = ZG'$. It is closed since $H = \det^{-1}((F^*)^2)$. It is cocompact since:

$$\frac{F^*}{(F^*)^2} \cong \frac{\mathbb{Z}}{2\mathbb{Z}} \times \frac{\mathcal{O}_F^*}{(\mathcal{O}_F^*)^2}$$

and $\mathcal{O}_F^*$ is compact and so is the quotient by a closed subgroup. The $\mathbb{F}_2$ -dimension of G/H is that of $F^*/(F^*)^2$ which is:

$$\dim_{\mathbb{F}_2} F^*/(F^*)^2 = \begin{cases} 2 + e & \text{if } \text{char}_F \neq 2, \\ \infty & \text{if } \text{char}_F = 2, \end{cases} \quad \text{where } 2\mathcal{O}_F = \mathfrak{P}_F^e.$$

Hence if $\text{char}_F \neq 2$, ZG' is open. Next, let $\bar{F}^{\text{sep}}$ represent the separable algebraic closure, $\mathcal{W}_F$ and G_F the Weil group and galois group of $\bar{F}^{\text{sep}}/F$ respectively. Let Ψ be a smooth additive K -character on F , trivial on $\mathcal{O}_F$ but not on $\mathfrak{p}_F^{-1}$ (i.e. a level 1 character, but not on $\mathfrak{p}_F^{-1}$ (i.e. A level 1 character).

Definition 3.1: Whittaker Space

The space of smooth K -characters of U have the form

$$\theta(u) = \psi \circ \text{tr}(Y(u - 1)) = \Psi(Y_{2,1}, u_{1,2}) \quad u \in U$$

for a lower triangular nilpotent matrix $Y \in M_2(F)$. The case $Y = 0$ gives the trivial character of U , the cases with $Y \neq 0$ give the *nondegenerate characters* of U .

When $Y_{2,1} = 1$, then [here](#), so denote $\theta_Y = \theta$.

[more words here before defining](#)

Definition 3.2: L-packet

Let Π be an irreducible smooth K -representation of G . Then the set $L(\Pi)$ of isomorphism classes of irreducible components of the restriction $\Pi|_{G'}$ is called an *L-packet*.

We classify the irreducible smooth K -representations of G' by these L-packets.

Proposition 3.3: Properties of L-packets: I

1. Any irreducible smooth K -representation of G' belongs to a unique L -packet.
2. For two irreducible smooth K -representations Π_1, Π_2 of G ,
$$L(\Pi_1) = L(\Pi_2) \iff \Pi_1 = (\chi \circ \det) \otimes \Pi_2$$
for some K -character $\chi \circ \det$ of G .
3. , The trivial character of G' is the unique finite-dimensional irreducible smooth K -representation of G' , it is degenerate and forms an L -packet $L(1) = L(\chi \circ \det)$ for any smooth K -character χ of F^* .
4. If Π is an irreducible smooth K -representation of G the restriction of Π to G' is semisimple of finite length and multiplicity 1.

Proof :
here

words here before

Proposition 3.4: Properties of L-packets: II

1. G_Π is the largest subgroup I of G such that $\text{len}(\Pi|_I) = \text{len}(\Pi|_{G'})$.
2. $\Pi = \text{ind}_{G_\Pi}^G V_\pi$ where V_π is the space of π .
3. $L(\Pi)$ is a principal homogeneous space for G/G_Π .
4. $|L(\Pi)|$ is a power of 2, and $|L(\Pi)|$ divides 2^{2+e} when $\text{char}_F \neq 2$.

Proof :
here

If p is odd, then $|F^*/(F^*)^2| = 4$, and hence:

Proposition 3.5: Size of L-packets

If p is odd, then the cardinality of an L -packet is either 1, 2, or 4.

Proof :
here

When $p = 2$, this more difficult as [HERE](#). This is proven via LLC.